# Chapter 3: Relational Database Design

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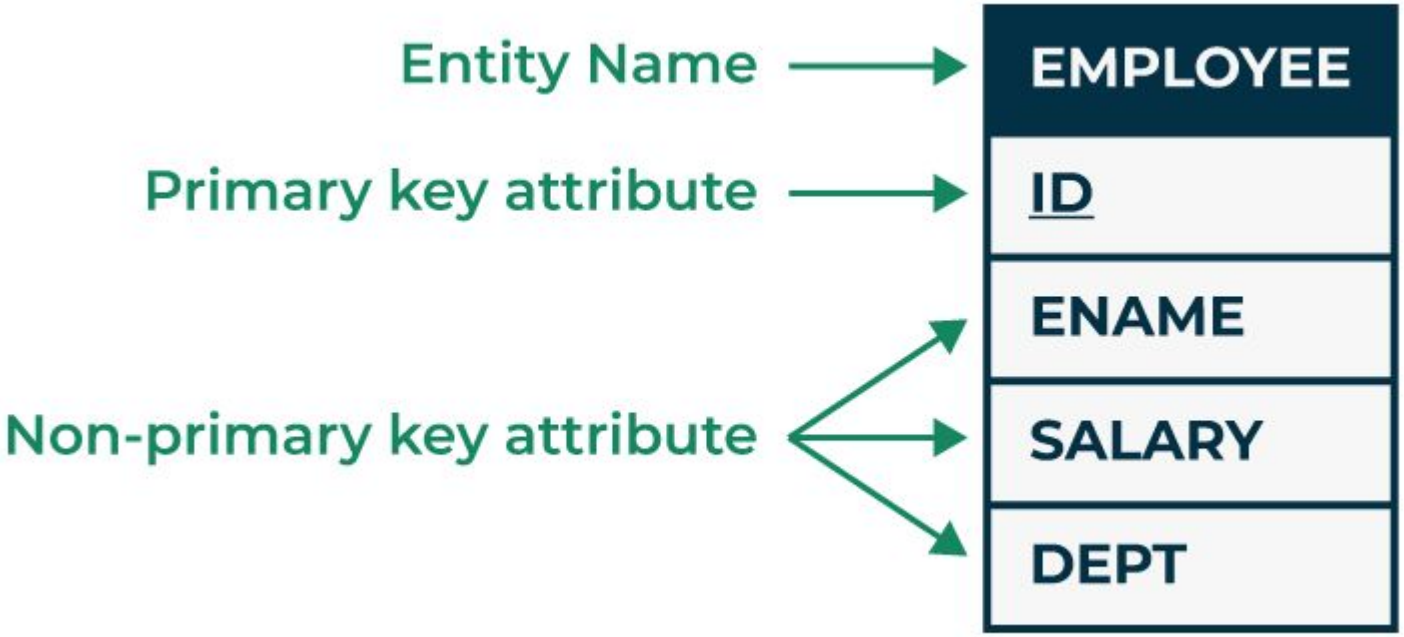
4. Purpose of Normalization.
5. Process of Normalization using: 1NF 2NF, 3NF.

# Entity Sets

- An entity is an object that exists and is distinguishable from other objects.
- Example: specific person, company, event, plant
- An entity set is a set of entities of the same type that share the same properties.
- Example: set of all persons, companies, trees,
- ▪ An entity is represented by a set of attributes; i.e., descriptive properties possessed by all members of an entity set.
- Example: instructor = (ID, name, salary ) course= (course_id, title, credits)

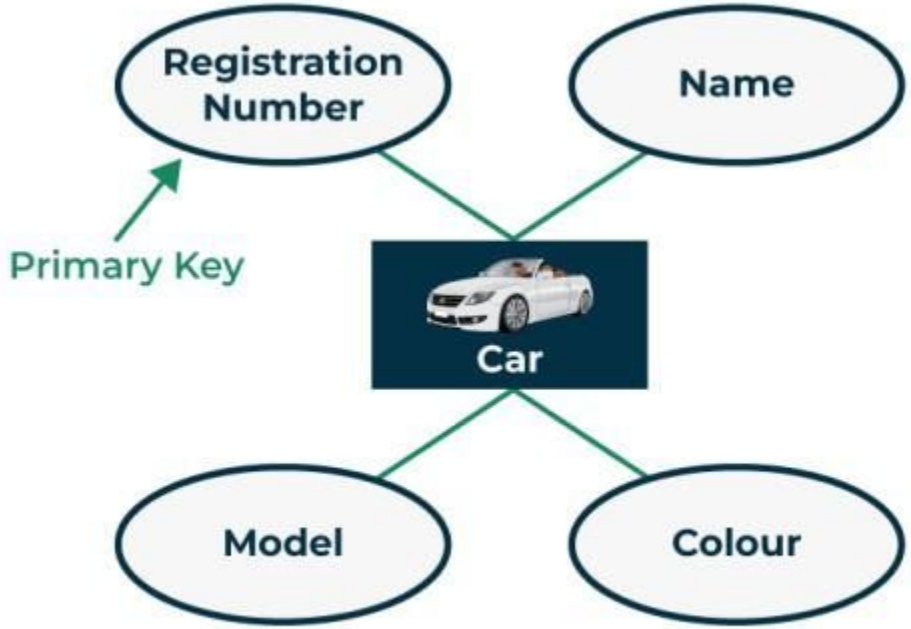
- An entity is a "**thing**" or "**object**" in the real world.
- An entity contains attributes, which describe that entity. So anything about which we store information is called an entity.



- If "John Doe" is an entity, then "Employees" is the entity set.
- **Key Characteristics**
- **Shared Attributes:** All entities within a set share the same properties (attributes). For example, every entity in a Student set will have a Name and an ID.
- **Primary Key:** Every entity set must have an attribute (or a combination of them) that uniquely identifies each entity within the set.
- **Mapping to Tables:** In a relational database, an **Entity Set** usually becomes a **Table**, and each **Entity** becomes a **Row** (tuple).

# Strong Entity Set

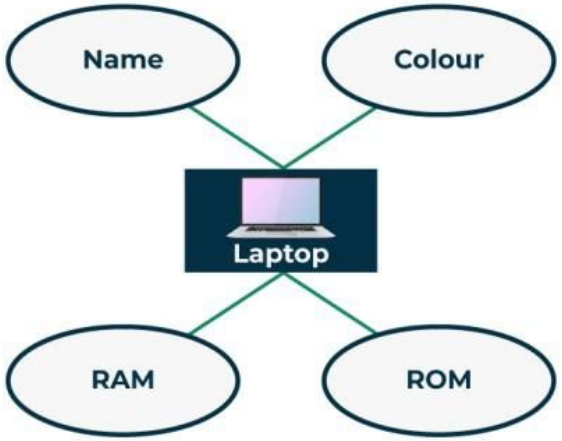
- Strong entity sets exist independently and each instance of a strong entity set has a unique primary key.
- Example of Strong Entity includes:
- Car Registration Number
- Model
- Name etc.



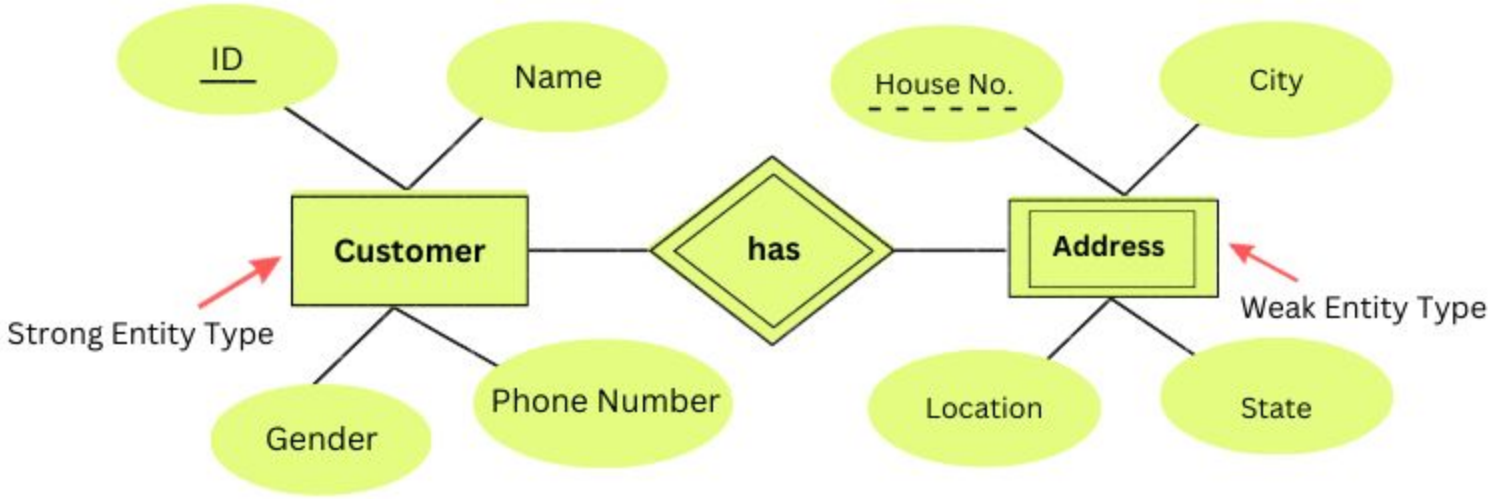

ER Diagram of Strong Entity Set

# Weak Entity Set

- A weak entity cannot exist on its own; it is dependent on a strong entity to identify it.
- A weak entity does not have a single primary key that uniquely identifies it; instead, it has a partial key.
- Example of Weak Entity Set includes:
- Laptop Color
- RAM, etc.

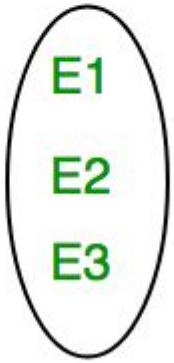



ER Diagram of Weak Entity Set




**Entity Relationship Diagram between Strong Entity Type & Weak Entity Type**

Entity Type




Entity Set

# What is Attributes?

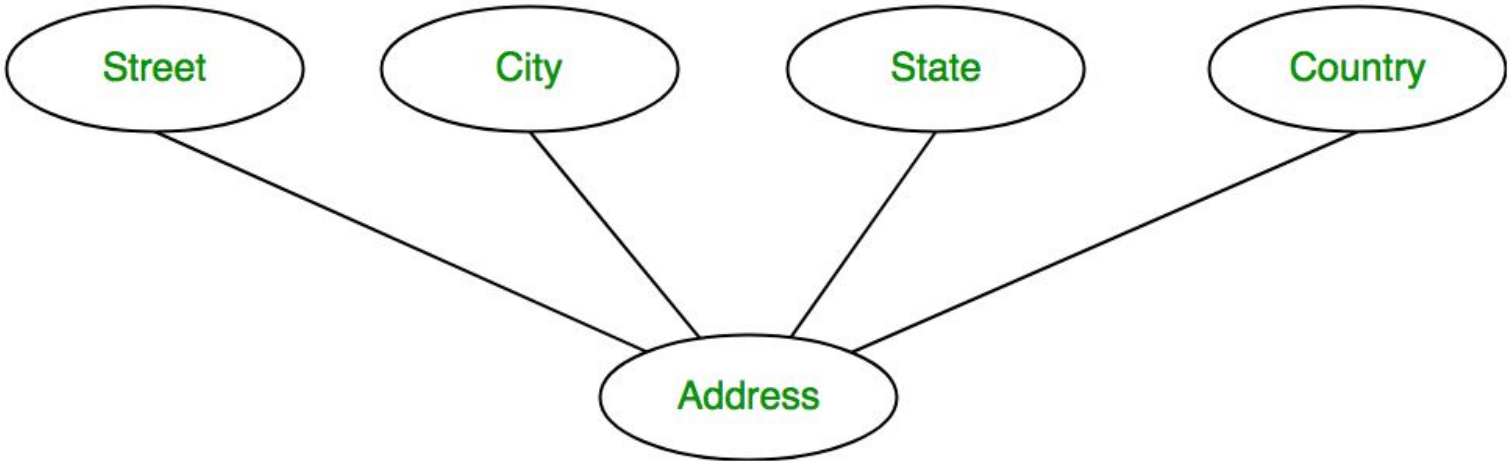
- properties that define the entity type. For example, Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.
- In ER diagram, the attribute is represented by an oval.

# Types of Attributes

**1.Key Attribute**

- The attribute which **uniquely identifies each entity** in the entity set is called the key attribute.
- For example, Roll_No will be unique for each student.
- In ER diagram, the key attribute is represented by an oval with underlying lines.

- **2. Composite Attribute**
- An attribute **composed of many other attributes** is called a composite attribute. For example, the Address attribute of the student Entity type consists of Street, City, State, and Country. In ER diagram, the composite attribute is represented by an oval comprising of ovals.



- **3. Multivalued Attribute**
- An attribute consisting of more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, a multivalued attribute is represented by a double oval.
-

- **4. Derived Attribute**
- an attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, the derived attribute is represented by a dashed oval.

# Relationship Type and Relationship Set

- A Relationship Type represents the association between entity types.
- For example, 'Enrolled in' is a relationship type that exists between entity type Student and Course.
- In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.

# Degree of a Relationship Set

- **1. Unary Relationship:** When there is only ONE entity set participating in a relation, the relationship is called a unary relationship.
- For example, one person is married to only one person.
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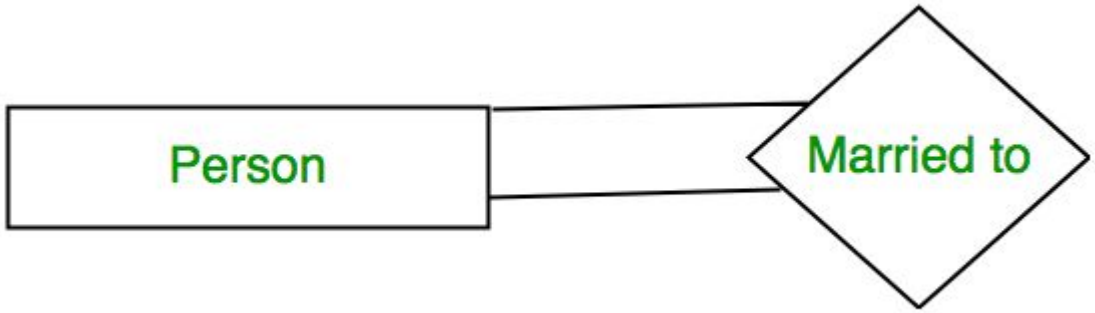


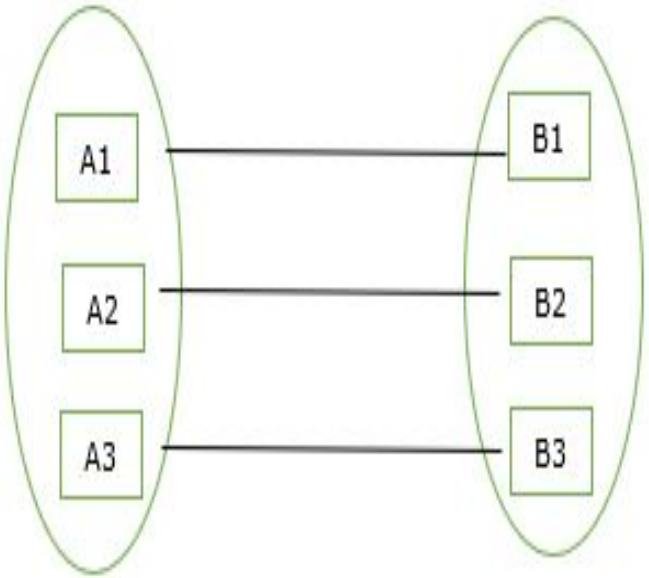
- **2. Binary Relationship:** When there are TWO entities set participating in a relationship, the relationship is called a binary relationship. For example, a Student is enrolled in a Course.
-

# Mapping Cardinality

- It is expressed as the number of entities to which another entity can be associated via a relationship set.
- Types of cardinality in between tables are:
- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many

# One-to-One

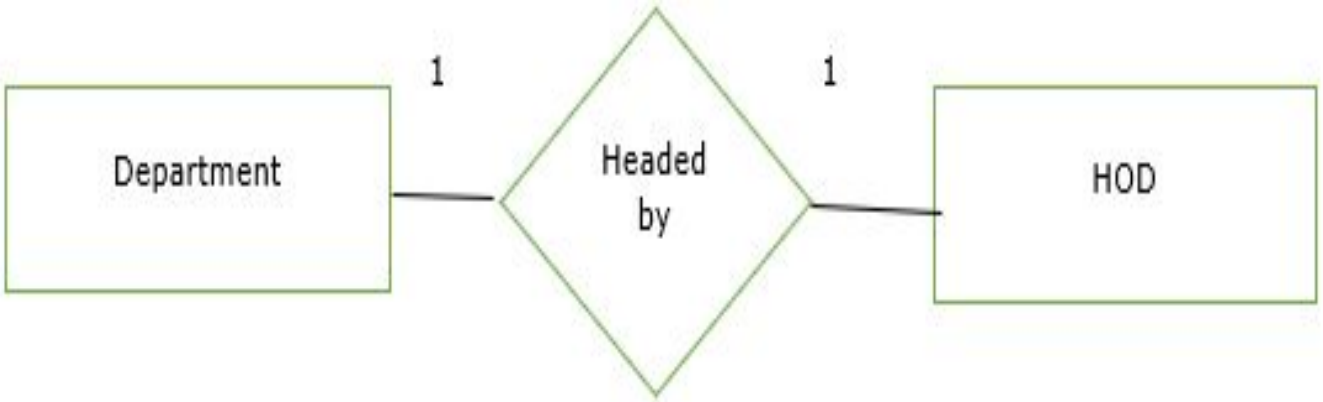
- In this type of cardinality mapping, an entity in A is connected to at most one entity in B or vice versa




**Example:**
**Country and Capital City.**
A country has only one capital.
A capital city belongs to only one country.

# Example

- Given below is an example of the one-to-one relationship in the mapping cardinality.
- Here, one department has one head of the department (HOD).



# One-to-Many

- In this type of cardinality mapping, an entity in A is associated with any number of entities in B. Or we can say that one unit or item in B can be connected to at most one unit or item in A.
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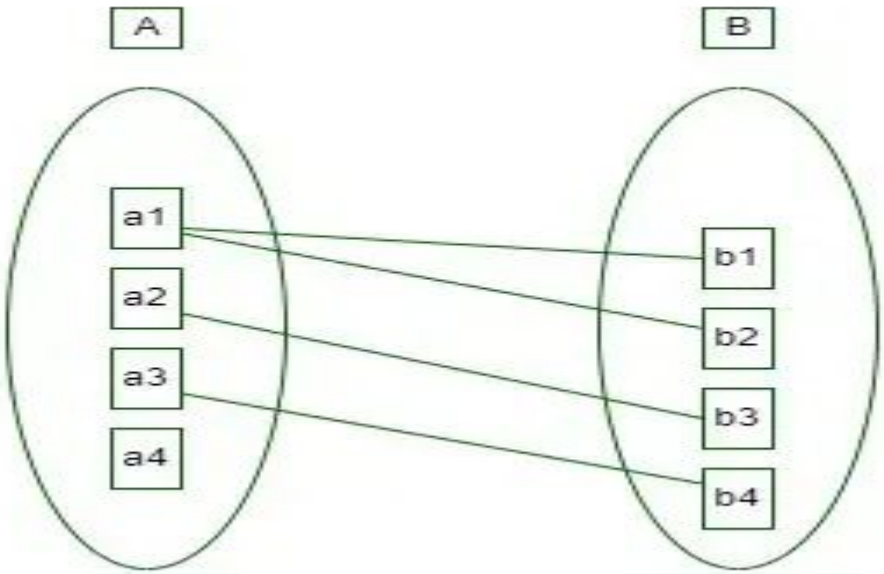
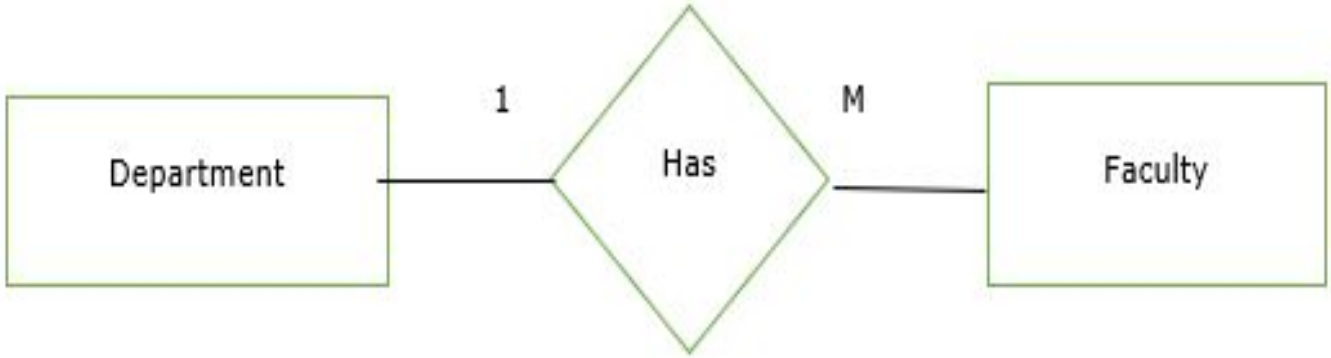


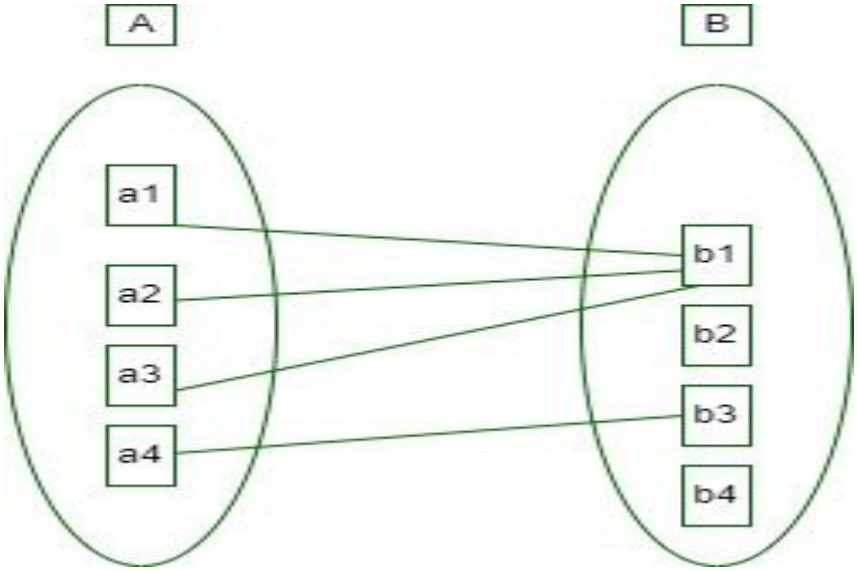
# Example

- Given below is an example of the one-to-many relationship in the mapping cardinality. Here, one department has many faculties.



# Many-to-one relationship

- In this type of cardinality mapping, an entity in A is connected to at most one entity in B. Or we can say a unit or item in B can be associated with any number (zero or more) of entities or items in A.

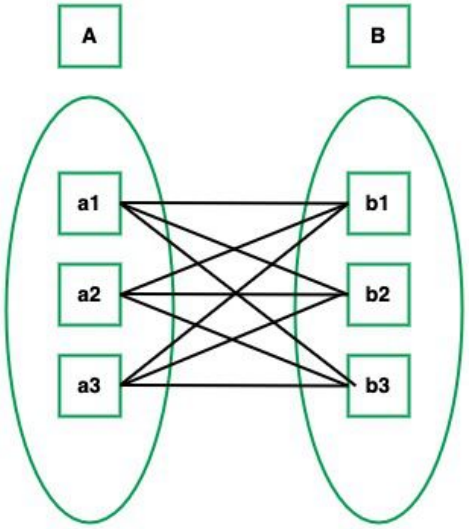


# Example

- Given below is an example of the many-to-one relationship in the mapping cardinality. Here, many faculties work in one department.

# Many-to-many relationship

- In this type of cardinality mapping, an entity in A is associated with any number of entities in B, and an entity in B is associated with any number of entities in A.

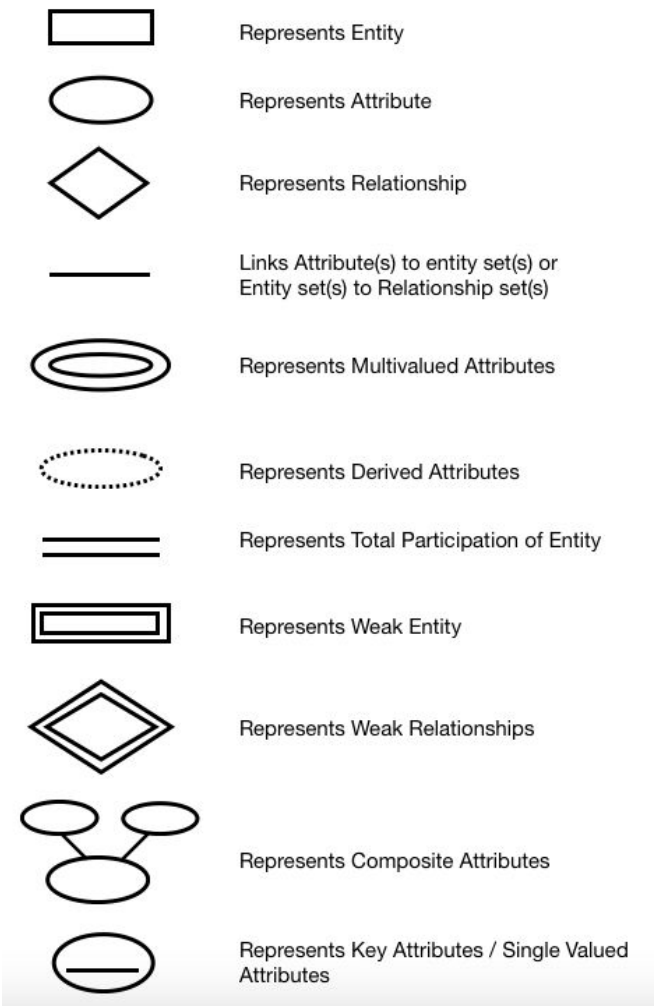


# Example

- Given below is an example of the many-to-many relationship in the mapping cardinality. Here, many employees work on many projects.

| Cardinality | Entity A → Entity B | Entity B → Entity A | Example |
|---|---|---|---|
| **1:1** | Max 1 | Max 1 | Person ↔ Passport |
| **1:N** | Many | Max 1 | Hospital ↔ Patient |
| **M:N** | Many | Many | Author ↔ Book |

# E-R Daigram Notations



# Extended ER (EER) features

- 1. Specialization (Top-Down Approach)
- 2. Generalization (Bottom-Up Approach)

# 1. Specialization (Top-Down Approach)

- Specialization is the process of breaking down a high-level entity into lower-level entities based on specific characteristics. It is a **top-down** approach.
- start with a general entity (Superclass) and create specialized versions of it (Subclasses) that have their own unique attributes.
- **Example:** Consider a Person entity. In a university, a person can be specialized into a Student or a Teacher.
- **Person (Superclass):** Attributes like Name, Address.
- **Student (Subclass):** Has unique attribute GPA.
- **Teacher (Subclass):** Has unique attribute Salary.

## 2. Generalization (Bottom-Up Approach)

- Generalization is the exact opposite of Specialization. It is a **bottom-up** approach where you take multiple entities with common features and combine them into a single, higher-level entity.
- You identify common attributes across different entities and "generalize" them.
- **Example:** Imagine you have an entity Car and an entity Truck.
- Both have a LicensePlate, EngineNumber, and Make.
- You can generalize them into a higher-level entity called Vehicle.